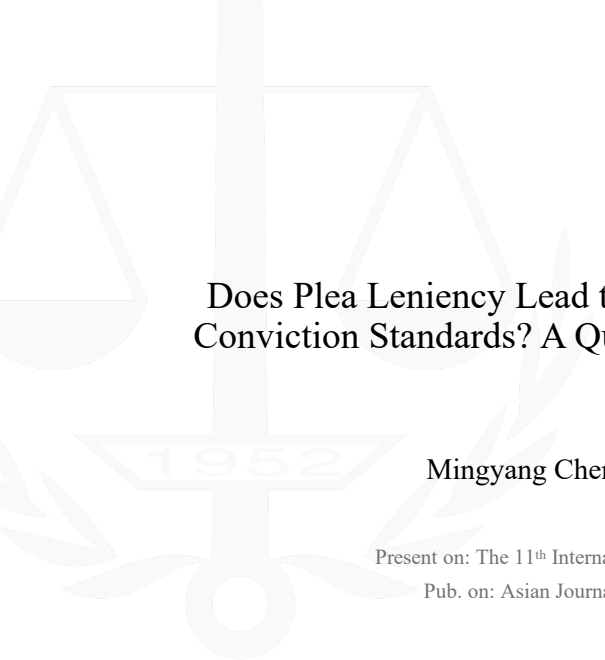




Does Plea Leniency Lead to Higher Crime Rates by Lowering Conviction Standards? A Quasi-Experimental Analysis in China

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BACKGROUND: Efficiency, Human Rights and Plea Leniency

全国人民代表大会常务委员会关于授权最高人民法院、最高人民检察院在部分地区开展刑事案件认罪认罚从宽制度试点工作的决定

发 文 机 关：全国人民代表大会常务委员会

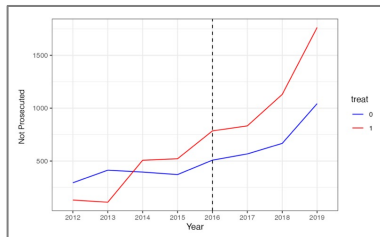
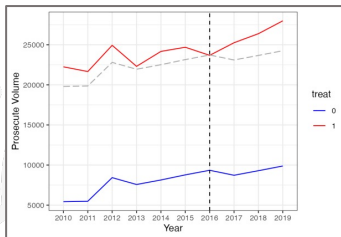
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时 效 性：现行有效

全国人民代表大会常务委员会关于授权最高人民法院、最高人民检察院在部分地区开展刑事案件认罪认罚从宽制度试点工作的决定

（2016年9月3日第十二届全国人民代表大会常务委员会第二十二次会议通过）



1. Efficiency and human rights protection often appear to be in conflict within the criminal justice system (Packer, 1964). Prosecutors lower conviction standard through plea bargain system and many innocent defendants are sentenced (Marcus, Brook, et al., 2016).
2. Chinese government asserts that this system offers a dual guarantee of both human rights and efficiency.
3. Descriptive results suggest that the Plea Leniency system is achieving its stated objectives.

LITERATURE REVIEW: Why Prosecution Volume Increases?

1. Efficiency:

1. Wu (2024) finds that Plea Leniency expedites criminal proceedings.
2. This acceleration is expected to result in a higher prosecution volume.

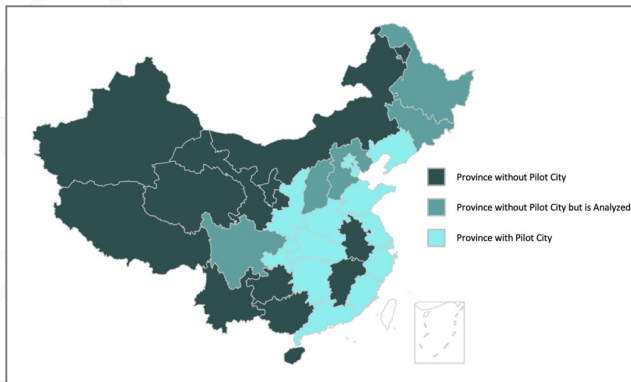
2. Deterrence:

1. Plea Leniency system contributes to a lower cost of crime by reducing overall punishment severity (Wu, 2020b, 2020c, 2024; Liu, 2021).
2. Beccaria (1872) argued that punishment serves to reduce crime.

3. Lower Conviction Standard:

1. The pressure faced by defendants compels them to plead guilty.
2. Prosecutorial success is measured by conviction rates, incentivizing a high rate of prosecutions. China's bureaucratic hierarchy aggravates it (Lin & Chen, 2020).

DATA: Sample



Sample: 95 cities in China covering the period from 2012 to 2019.

DATA: Variables

Table 2. Statistics of Variables

Statistic	Observations	Mean	St. Dev.	Min	Max
<i>Dependent Variables:</i>					
Crimerate_Filed	457	49.777	32.140	8.360	220.480
Crimerate_Solved	509	21.279	13.839	4.150	154.610
Crimerate_Prosecuted	711	11.881	5.212	3.190	44.350
Crimerate_Arrested	675	7.439	4.130	1.680	39.070
Crimerate_Uploaded	380	4.950	2.455	1.630	18.160
Crimerate_Concluded	691	10.527	7.056	1.190	71.680
Not_Prosecuted	471	0.768	0.835	0.020	6.460
Not_Arrested	477	1.653	1.237	0.110	7.710
Crimerate_Dealt	465	13.043	5.903	3.640	46.010
Prosecution_Rate	467	94.538	4.096	79.730	99.750
Arrest_Rate	467	83.141	7.422	61.450	98.110
<i>Control Variables:</i>					
GDP	760	5,107.240	7,236.702	236.140	72,609.910
GDP per capita	760	74,933.040	111,52.050	11,968.140	1,397,44.600
Unemployment Rate	760	2.779	0.734	1.200	5.000
Urbanization Ratio	760	67.368	75.668	23.020	820.000
Population	760	626.930	496.409	68.720	3,188.000
Population Growth	760	1.002	0.031	0.792	1.292
Population Density	760	805.334	799.909	88.720	6,728.000
Immigration	760	120.455	294.930	2.810	3,332.260
Sex Ratio	760	103.665	5.548	92.940	149.000
Education	760	239.696	242.675	15.160	1,100.670
URID	760	47.863	9.754	21.890	100.000
Urban Income	760	33,362.340	11,526.810	15,457.000	103,108.000
Rural Income	760	16,014.340	6,821.611	4,566.000	62,522.000
CPI	760	102.218	1.355	95.900	116.800
Age Structure	760	11.552	2.994	2.070	22.050
PSE	760	40.918	61.826	1.740	516.420
PSE per capita	760	558.404	334.895	30.750	2,355.930
PSE-GDP Ratio	760	90.356	37.664	5.610	269.030

Control Variables Note: URID means urban-rural income disparity; PSE means public safety expenditures; All units are illustrated in the article.

DATA: Quality of Our Data

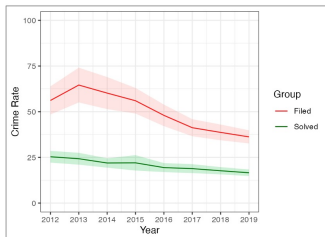


Figure 8. Statistics of Crimerate_Filed and Crimerate_Solved

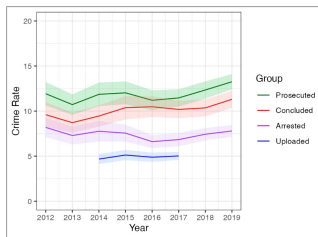


Figure 9. Statistics of Crimerate_Prosecuted, _Concluded, _Arrested and _Uploaded

Group	City-Level Mean	Nation-Level Mean	P-value
Crimerate_Prosecuted	11.881	11.179	0.176
Crimerate_Arrested	7.439	7.020	0.168
Crimerate_Filed	49.777	44.233	0.182
Crimerate_Concluded	10.527	9.354	0.048
Crimerate_Uploaded	9.901	9.353	0.267

Note: Crimerate_Uploaded represents the data after adjusting for a disclosure rate of 50%.

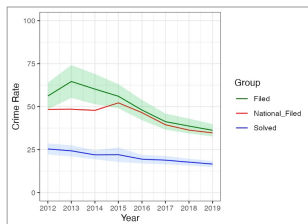


Figure 10. Statistics of City-Level and Nation-Level Police Data

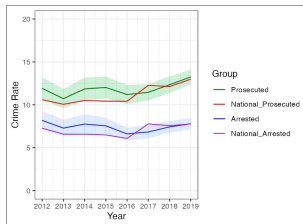


Figure 11. Statistics of City-Level and Nation-Level Procuratorate Data

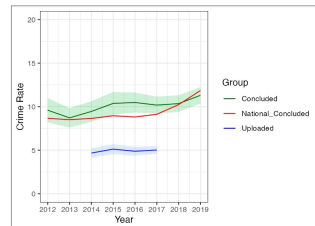


Figure 12. Statistics of City-Level and Nation-Level Court Data

EMPIRICAL STRATEGIES: DiD Model

Core Regression Models:

$$Crimerate_Prosecuted_{c,t} = \beta \cdot Treat_c \cdot Post2016_t + \gamma \cdot Control_{c,t} + \delta_c + \eta_t + \varepsilon_{c,t} \quad (1)$$

Mechanism Analysis Models:

$$Mechanism_{c,t} = \beta_0 \cdot D_{c,t} + \gamma_0 \cdot Control_{c,t} + \delta_{0c} + \eta_{0t} + \varepsilon_{0c,t} \quad (6)$$

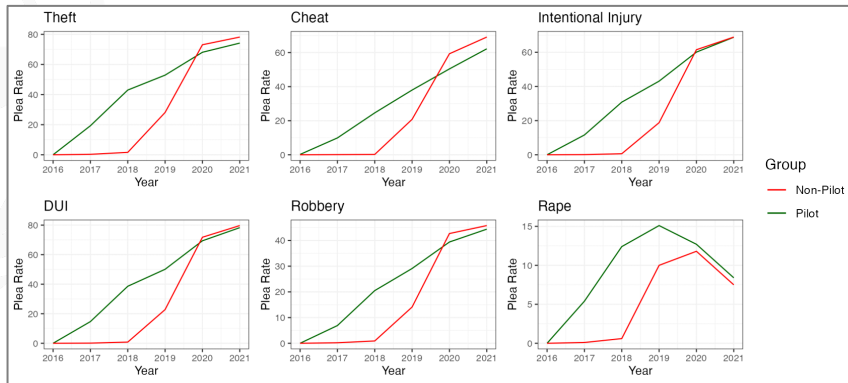
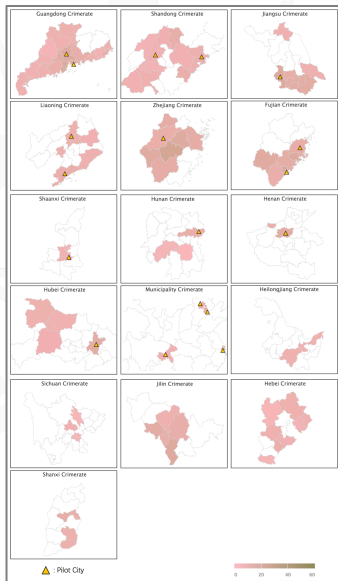
$$Crimerate_Prosecuted_{c,t} = \beta_1 \cdot D_{c,t} + \mu_0 \cdot Mechanism_{c,t} + \gamma_1 \cdot Control_{c,t} + \delta_{1c} + \eta_{1t} + \varepsilon_{1c,t} \quad (7)$$

$$Crimerate_Prosecuted_{c,t} = \mu_1 \cdot Mechanism_{c,t} + \gamma_2 \cdot Control_{c,t} + \delta_{2c} + \eta_{2t} + \varepsilon_{2c,t} \quad (8)$$

Non-Pilot Effect Models:

$$Crimerate_Prosecuted_{c,t} = \beta \cdot Anti_Treat_c \cdot Pre2018_t + \gamma \cdot Control_{c,t} + \delta_c + \eta_t + \varepsilon_{c,t} \quad (10)$$

EMPIRICAL STRATEGIES: Non-Pilot Effect (Wang, Yang, 2021)



RESULT: Core Regression Results

Table 3. Core Regression Results

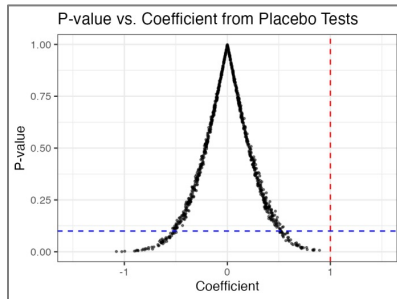
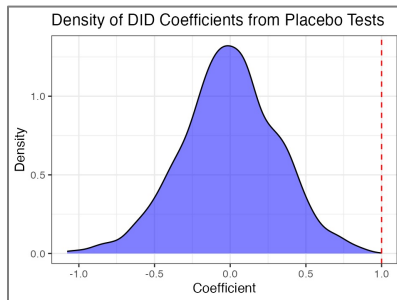
	<i>Dependent variable:</i>					
	Crimerate_Prosecuted		Crimerate_Arrested		Crimerate_Uploaded	
	(1)	(2)	(3)	(4)	(5)	(6)
Treat·Post2016	1.939** (0.763)	0.976* (0.512)	1.129* (0.636)	1.123*** (0.383)	0.737* (0.429)	0.694** (0.327)
Constant	11.513*** (0.227)		7.289*** (0.184)		4.880*** (0.132)	
Region FE	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes
Control	No	Yes	No	Yes	No	Yes
Observations	617	617	586	586	380	380
R ²	0.011	0.984	0.005	0.983	0.008	0.980
Adjusted R ²	0.009	0.980	0.004	0.980	0.005	0.971

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

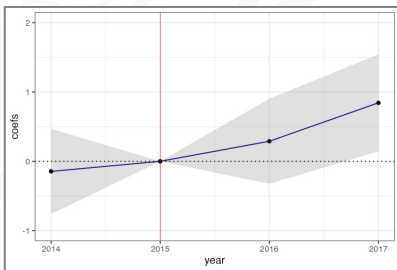
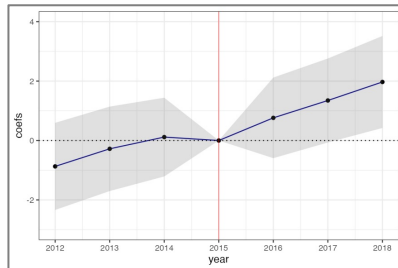
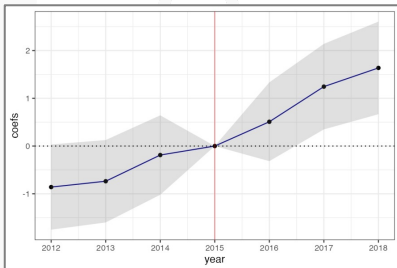
RESULT: Robustness Check

Table 4. Robustness Check for Core Regression

Dependent variable: Crimerate_Prosecuted										
Method:										
	Pre-Trend		Event Study		Placebo Test		GLM	Control Car	Fixed Baseline Population	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treat_Post2016							0.090** (0.035)	1.215** (0.553)	0.960* (0.548)	0.903* (0.504)
Treat_Post2015					0.369 (0.499)					
Treat_Post2014						0.603 (0.502)				
Treat_Year2012	-1.601** (0.745)		-0.792 (0.760)							
Treat_Year2013	-1.201* (0.719)		-0.361 (0.709)	0.431 (0.723)						
Treat_Year2014	-0.622 (0.625)		0.267 (0.704)	1.060 (0.752)						
Treat_Year2015	-0.963 (0.620)			0.793 (0.756)						
Treat_Year2016		0.436 (0.596)	0.292 (0.704)	1.122 (0.791)						
Treat_Year2017		1.525** (0.655)	1.470** (0.766)	2.263*** (0.859)						
Treat_Year2018		1.558** (0.737)	1.557** (0.796)	2.350** (0.951)						
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes



RESULT: Robustness Check



From left to right, up to down:
Event Study of $Crimerate_Arrested_{c,t}$
Event Study of $Crimerate_Concluded_{c,t}$
Event Study of $Crimerate_Uploaded_{c,t}$

RESULT: Mechanism Analysis

Table 9. Prosecution Rate Effect

	<i>Dependent variable:</i>										
	Not_Prosecuted		Prosecution_Rate		Not_Arrested		Arrest_Rate		Robustness Check		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Treat·Post2016	0.175 (0.134)	-0.440*** (0.142)	0.076 (0.663)	2.671*** (0.793)	1.261*** (0.206)	-0.332 (0.213)	-4.759*** (1.277)	3.308 (2.012)	0.104** (0.043)	0.044* (0.024)	0.048** (0.020)
Constant	0.668*** (0.040)		94.957*** (0.197)		1.504*** (0.061)		83.837*** (0.372)				
Region FE	No	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes
Control	No	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes
Observations	413	413	409	409	420	420	413	413	378	368	232
R ²	0.004	0.907	0.00003	0.999	0.083	0.945	0.033	0.997	0.933	0.956	0.958
Adjusted R ²	0.002	0.870	-0.002	0.999	0.080	0.925	0.030	0.996	0.909	0.940	0.929

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

RESULT: Mechanism Analysis

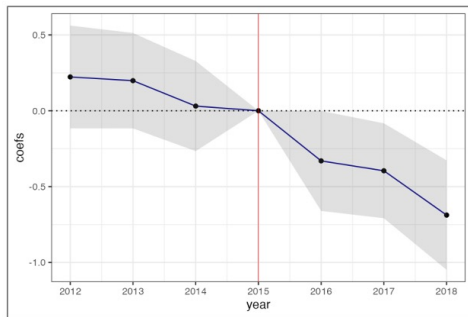


Figure 25. Event Study of $Not_Prosecuted_{c,t}$

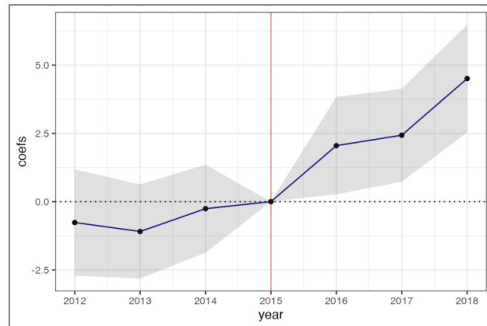


Figure 26. Event Study of $Prosecution_Rate_{c,t}$

RESULT: Mechanism Analysis

Table 11. Further Mechanism Check

	<i>Dependent variable:</i>							
	Crimerate_Prosecuted							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treat·Post2016		0.369*** (0.126)		1.989** (0.782)		1.149** (0.652)		0.654 (0.720)
Crimerate_Dealt	0.959*** (0.010)	0.958*** (0.010)						
Crimerate_Filed			0.039*** (0.008)	0.041*** (0.008)				
Crimerate_Solved					0.025** (0.012)	0.024*** (0.012)		
Prosecution_Rate							0.137*** (0.052)	0.128** (0.053)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	407	407	378	378	444	444	405	405
R ²	0.999	0.999	0.986	0.987	0.985	0.985	0.987	0.987
Adjusted R ²	0.999	0.999	0.982	0.982	0.980	0.980	0.981	0.982

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

RESULT: Non-Pilot Effect

Table 12. Pilot Selection Problem

	Dependent variable:				
	Treat (0,1)				
	(1)	(2)	(3)	(4)	(5)
Crimerate_Filed	0.017*** (0.005)				
Crimerate_Prosecuted		0.084*** (0.022)			
Crimerate_Arrested			0.106** (0.027)		
GDP				0.001*** (0.000)	
Urbanization					0.084*** (0.010)
(Intercept)	-2.718*** (0.382)	-2.382*** (0.317)	-2.238*** (0.275)	-4.271 (0.000)	-6.922*** (0.722)
Observations	205	337	324	380	380
R ²	0.513	0.108	0.172	0.491	0.292
Adjusted R ²	0.507	0.103	0.167	0.486	0.287
RMSE	0.38	0.40	0.40	0.28	0.34

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 13. Non-Pilot Effect

	Dependent variable:					
	Crimerate_Prosecuted			Prosecution_Rate		
	(1)	(2)	(3)	(4)	(5)	(6)
Anti_Treat·Pre2018	0.681 (0.584)			0.410 (1.142)		
Treat·Post2016		0.715* (0.490)	0.968* (0.532)		2.45*** (0.773)	2.466*** (0.841)
2012-2018 Coef.		0.976* (0.512)	1.215** (0.553)		2.671*** (0.793)	2.671*** (0.817)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Control	Yes	Yes	Yes	Yes	Yes	Yes
Control Car_Per	No	No	Yes	No	No	Yes
Observations	366	703	621	236	467	418
R ²	0.990	0.983	0.984	1.000	1.000	1.000
Adjusted R ²	0.985	0.979	0.980	1.000	0.999	0.999

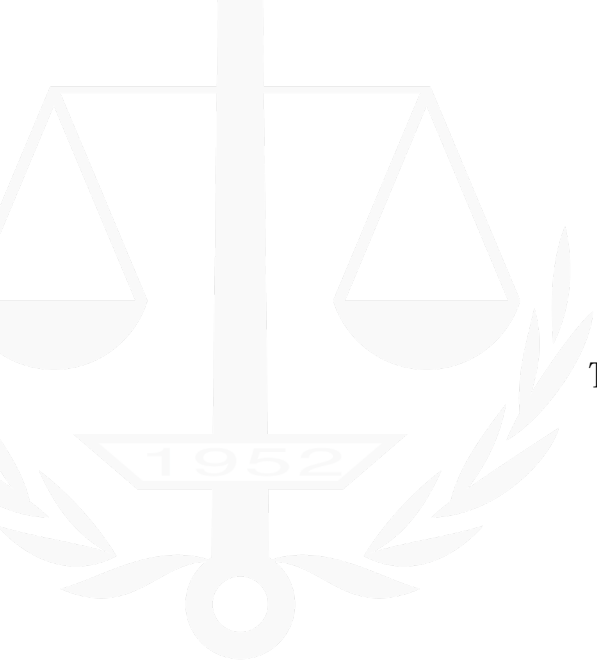
Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Conclusion

1. China's approach has not fully resolved the conflict between efficiency and human rights inherent in the plea bargaining system.
2. The Plea Leniency system places excessive emphasis on speed and efficiency.
3. China's criminal justice system is moving further away from ensuring the protection of human rights.

认罪认罚成效显著江西新闻客户端讯(江西日报全媒体记者朱雪军 通讯员韦明月) 今年以来, 庐山市人民检察院加大认罪认罚工作力度, 适用率稳步提高, 由2024年度的87.84%提升至100%。在审查逮捕、起诉时, 同步开展认罪认罚告知制度, 教育引导犯罪嫌疑人真诚认罪悔罪, 确保认罪认罚自愿真实合法。

We successfully predict what will happen in the near future



THANK YOU